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Structural and magnetic properties of NiS doped Bi-2212 superconductors

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Abstract

In this work we have studied the effect of NiS doping on the superconducting properties of the Bi-2212 (nominal composition $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$), superconductors. Our studies show that NiS doping stabilizes superconductivity in the system but all the NiS doped samples gave a T_c lower than the superconducting transition temperature of pure Bi-2212 superconductors. It appears that NiS substitution for CuO planes and modification of the oxygen content due to S substitution on O places jointly produce strong localization of charge carriers and at the same time increase the hole concentration of the system. These combined phenomena occurring in the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ system are responsible for the deterioration of superconductivity of NiS doped Bi-2212 superconductors. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$; NiS doping; Bi-2212 superconductors

1. Introduction

In high temperature superconductors (HTS) substitution and doping studies are of immense importance for basic as well as applied aspects. Due to the existence of essentially one kind of Cu site, the $\text{Bi}_2\text{O}_3\text{--SrO--CaO--CuO}$ (BSCCO) superconductor system is a simpler candidate for studying the effect of substitution and doping on the superconducting properties [1,2]. For on-site Cu substitution, the 3d metallic elements possess a favorable condition due to their close ionic sizes and outer shell configurations to that of copper. Normally 3d-metal oxides are used for substitution

and doping studies in HTS. In a recent study, Albino Aguiar et al. [3] observed that in $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ superconductors metal sulfide doping is less effective in deteriorating the superconducting transition temperature than metal oxide doping. Using a similar approach, in a recent study, we have successfully achieved stabilization and substantial increase in T_c of Bi-2212 superconductors using CuS doping [4].

Extending the above studies, in the present work, we have studied the effect of the NiS doping on the superconducting properties of the Bi-2212 (nominal composition $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$) phase. Hexagonal NiS exhibits a phase transition when the temperature or pressure is increased above 265 K and 20 GPa, respectively. NiS passes from an antiferromagnetic

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semiconducting state at <265 K to a Pauli paramagnetic metal state at >265 K. Transition to the antiferromagnetic state is accompanied by a 2% volume increase due to the lattice expansion [5]. In this way the double chemical substitution of Ni for Cu together with S or O can take place in the Bi-2212 structure, however EDX data are needed for this observation.

This article reports X-ray diffractometry studies and magnetization as a function of NiS substitution and discusses the decrease in T_c due to NiS doping in Bi-2212 superconductors.

2. Experimental

NiS doped Bi-2212 polycrystalline ceramics with nominal composition $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$, ($0 < x < 0.1$) were prepared by solid state reaction process. Constituent chemicals Bi_2O_3 , SrCO_3 , CaCO_3 , CuO and NiS (purity 99.99%) were mixed in the desired stoichiometric ratio and were dried in an oven at 200°C for 6 h. The dried mixture was finely powdered in an agate mortar and pelletized as circular discs, at a pressure of 2 ton/cm². The pelletized samples were calcined at 800°C for 40 h in a microprocessor controlled tube furnace. After calcination, samples were cooled to 500°C at a rate of $1^\circ\text{C}/\text{min}$ and then furnace cooled to room temperature. The calcined material was finely reground and pelletized as circular discs at a pressure of 2 ton/cm². Final heat treatment of $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ was carried out at 850°C for 100 h in air atmosphere. The heat treated samples were cooled at $1^\circ\text{C}/\text{min}$ to 500°C for proper oxygenation then furnace cooled to room temperature. All the sample processing was carried out in air. X-ray diffraction (XRD) spectra of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples were recorded at room temperature by a Siemens D-5000 X-ray diffractometer using CuK_α radiation ($\lambda = 1.5406 \text{ \AA}$). Superconducting properties of the samples were studied by measuring dc magnetization of the samples, in zero field cooled (ZFC) and field cooled (FC) conditions in the temperature range 5–300 K, at an applied dc magnetic field of 5 Oe using a Quantum Design (MPMS-5S) SQUID magnetometer.

3. Results and discussion

The XRD spectra of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples are shown in Fig. 1. From an analysis of the XRD peaks of $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples we conclude that all the samples have a typical crystal structure of Bi-2212 superconductors. In the XRD spectra of the samples, there are no reflections due to Bi-2201 and Bi-2223 superconducting phases. Although the XRD spectrum of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ sample with $x = 0.01$ NiS doping is not so smooth as of the other samples, all the samples are free from any other impurity phases. All the XRD peaks were identified and indexed for the single phase Bi-2212 superconductors.

The temperature dependence of the dc magnetization (FC and ZFC) of $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples in the temperature range 5–300 K is shown in Fig. 2. All the samples show superconducting behavior in the measured temperature range. No superconducting transition at 110 and 10 K, characteristics of the Bi-2223 and Bi-2201 phases respectively, are detectable in any sample. A saturated diamagnetic transition is clearly observed in every sample at temperatures below T_c . The only discrepancy observed in the magnetic properties is the ZFC magnetization curve of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ sample with $x = 0.01$ of NiS doping. The reason behind this is the fact that it is very difficult to dope the Bi-2212 samples with such a small doping

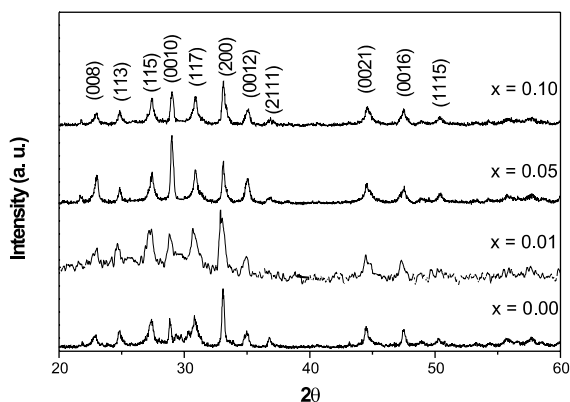


Fig. 1. XRD spectra of $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ superconductors.

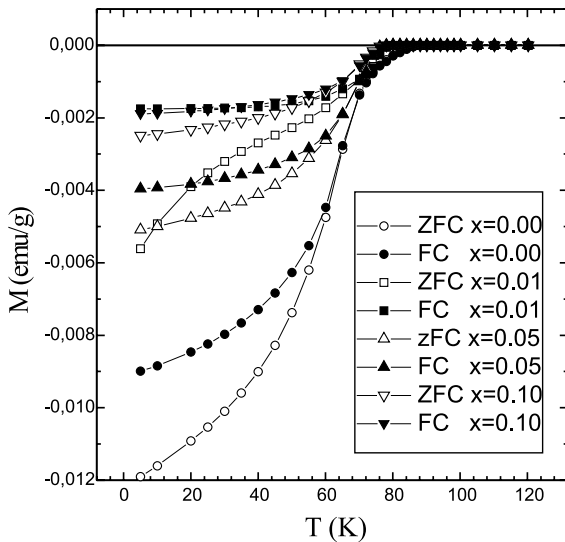


Fig. 2. dc magnetization, M versus temperature curves of $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ superconductors (ZFC – zero-field cooled and FC – field cooled).

level. These observations provide a clear evidence of the stabilization of Bi-2212 superconducting phase in the NiS doped $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples. A detailed view of the superconducting transition region is shown in Fig. 3. The $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ sample without NiS substitution gave the same T_c (88 K) as that of the single phase Bi-2212 superconductor [6]. The NiS doping in $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ decreases the superconducting transition temperature of the samples considerably. All the NiS doped $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples gave lower T_c than that of the pure sample. The doping dependence of the superconducting transition temperatures T_c , defined as the onset of the diamagnetic transition, of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ samples, obtained from the dc magnetization versus temperature measurements, is plotted in Fig. 4.

In recent studies on the phase diagram of the $\text{Bi}_2\text{O}_3\text{--SrO--CuO}$ system [7] it is found that several phases occurring in this system have variable oxygen stoichiometry as a function of temperature and oxygen activity in the surrounding atmosphere. Consequently the respective phase equilibria are dependent on the partial pressure of oxygen. In Bi-2212 superconductors, the transition

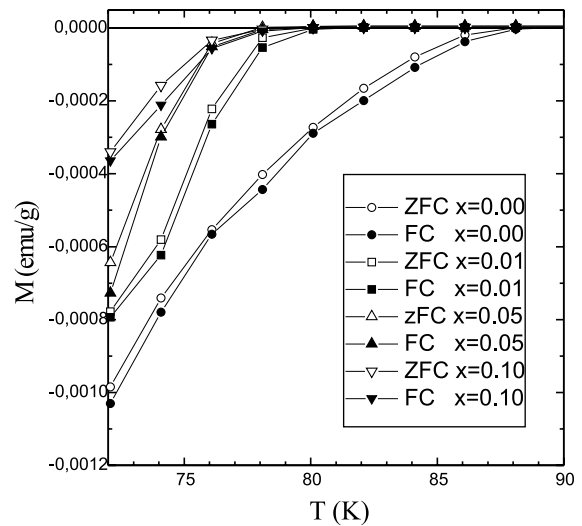


Fig. 3. Detailed view of the superconducting transition region of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ superconductors in Fig. 2.

to superconductivity depends on the hole concentration of the material as seen in YBCO superconductors, with the threshold value of 0.2 holes/Cu ions [8]. This change in the hole concentration can produce different T_c due to different oxygen stoichiometries. In YBCO superconductors doped with NiO, Ni normally substitutes on the Cu(II) site [9], while YBCO superconductors doped with NiS, NiS substitutes for Cu(I)O isostructurally, at least up to 10 at.% of NiS doping level [3]. Due to this, NiS doping in YBCO has less effect in deteriorating T_c than NiO doping.

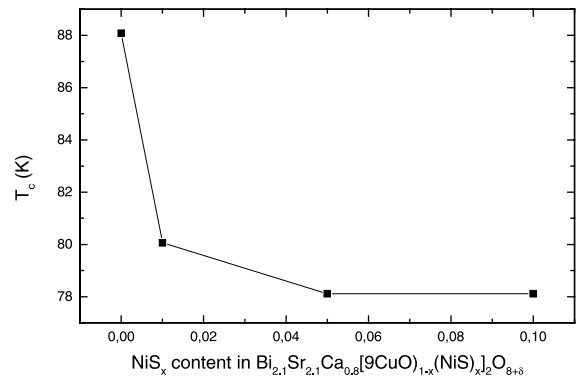


Fig. 4. Variation of transition temperature T_c with NiS_x content in $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ superconductors.

As seen from the XRD spectra of the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ system, there are no impurity peaks or reflections which could be assigned to unreacted NiS. Therefore it may be inferred that NiS is successfully doped in the Bi-2212 lattice. Due to the existence of essentially one kind of Cu site in Bi-2212 superconductors, there is a strong possibility that NiS substitutes for CuO isostructurally in the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ system. In this way there will be strong localization of charge carriers in the CuO planes and it will affect T_c anomalously and decrease the superconducting transition temperature. The magnetic moment of Ni does not appear to affect the superconductivity.

In our study on CuS doping in Bi-2212 superconductors, we observed that S substitutes for O, although very likely there are fewer S on O sites than the nominal composition of the starting powders suggests. The doping improves the oxygen content and increases in T_c of the CuS doped Bi-2212 superconductors [4]. Majewski and Hettich [10] showed that the critical temperature in sintered Bi-2212 reaches a maximum of 95 K and depends on the cation ratio as well as the oxygen content of the material. The origin of this double dependence lies in the hole concentration on the Cu–O planes which determine the critical temperature. An interesting observation of T_c versus the NiS concentration curve in Fig. 4 is that there is initially a rapid decrease in T_c with increasing NiS content that becomes more or less constant.

In the present study, it appears that NiS substitution on the CuO planes and modification of oxygen content due to S substitution on O places jointly produce strong localization of charge carriers and at the same time increase the hole concentration of the system. We conclude that combined phenomena occurring in $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ system are responsible for small amount of the deterioration of superconductivity in NiS doped Bi-2212 superconductors.

4. Conclusion

Our studies on NiS doping in Bi-2212 with nominal composition $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$

show that NiS doping stabilizes superconductivity in the system but all the NiS doped samples gave a T_c lower than undoped Bi-2212. It appears that NiS substitution on CuO planes and modification of the oxygen content due to S substitution on O places jointly produce strong localization of charge carriers and at the same time increase the hole concentration of the system. These combined phenomena occurring in the $\text{Bi}_{2.1}\text{Sr}_{2.1}\text{Ca}_{0.8}[(\text{CuO})_{1-x}(\text{NiS})_x]_2\text{O}_{8+\delta}$ system are responsible for the small amount of deterioration of superconductivity in NiS doped Bi-2212 superconductors.

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